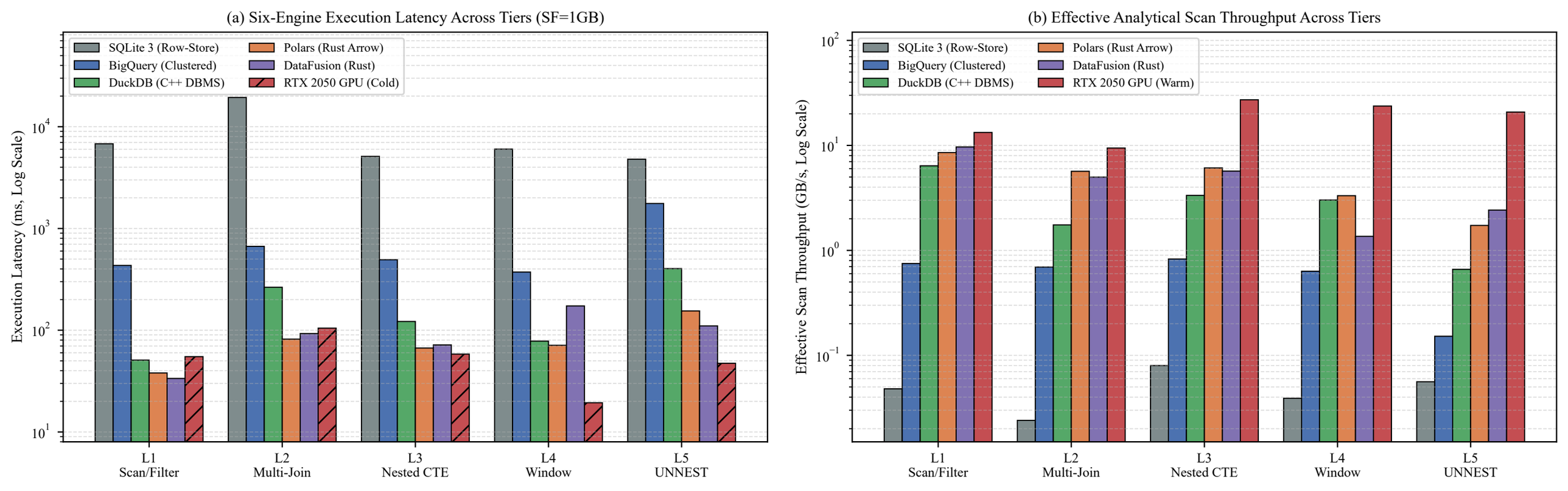
(a) Six-Engine Execution Latency Across Tiers (SF=1GB)
Execution Latency (ms, Log Scale)
SQLite 3 (Row-Store)
BigQuery (Clustered)
DuckDB (C++ DBMS)
Polars (Rust Arrow)
DataFusion (Rust)
RTX 2050 GPU (Cold)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(b) Effective Analytical Scan Throughput Across Tiers
Effective Scan Throughput (GB/s, Log Scale)
SQLite 3 (Row-Store)
BigQuery (Clustered)
DuckDB (C++ DBMS)
Polars (Rust Arrow)
DataFusion (Rust)
RTX 2050 GPU (Warm)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST